

The Simpler Competitor

Whole blood and dried plasma are winning the race that BHOC was built to run, and the honest map of where a cow protein still matters

The standard pitch for an oxygen carrier goes like this: blood is scarce, perishable and dangerous, so give the patient something better. It is a pitch that made sense in 1985, when the blood supply had been shaken by a new virus and by a shortage of donors in a war zone. In 2026 it needs a more careful hearing, because the thing it claims to beat has changed. Blood in the field is getting simpler, closer to the patient and better studied. The competitor is no longer a hospital blood bank a helicopter ride away. It is increasingly a cooler in the back of an ambulance.

This essay is a stress test. It asks, with as little wishful thinking as possible, where a BHOC still has a defensible place given what the simpler alternatives can now do. I believe the answer is real and narrower than promoters suggest, and that a patient organization gains more from stating the narrow answer plainly than from repeating a broad one that experienced clinicians will pick apart.

The competitor learned from the war

In the wars of the early twenty-first century, military medicine relearned an old lesson. Patients who die of hemorrhage die early, often within the first hour, and the intervention that changes their odds is giving blood products promptly and in something like the proportions in which they were lost. Whole blood, which contains red cells, plasma and platelets together, fits that principle naturally.

Two developments then moved whole blood from the battlefield to the civilian ambulance. First, the concept of low-titer group O whole blood took hold. Group O red cells can be given to almost anyone. Group O plasma contains antibodies against A and B antigens, which can harm recipients of other groups, but if the antibody level is low the risk is small. Blood banks began screening group O donors for low antibody titers, with a common threshold in the trial described below being a titer below 256, and issuing that blood as a near-universal product. Second, small teams in helicopters and ground ambulances began carrying it, along with portable refrigerators.

Meanwhile, plasma moved forward on a parallel track. A large randomized trial of plasma given during air medical transport, known as PAMPer and published in 2018, found lower mortality at 30 days in trauma patients who received plasma early. In the trial's control group, mortality was about 33 percent, and in the plasma group roughly 23 percent. A post hoc analysis in a later paper suggested that the benefit was concentrated in patients whose transport time was longer than about 20 minutes. Dried plasma, which can be stored at room temperature and reconstituted with sterile water, offered a way to bring the same principle to places without refrigeration.

The trial that complicated the story

The most recent large study of prehospital whole blood, called TOWAR, was published in the New England Journal of Medicine in May 2026. It enrolled 1,020 patients across 44 air medical bases and 11 trauma centers, between May 2022 and June 2025, using a design in which bases were randomly assigned, two to one, to carry whole blood or the usual component therapy. The primary outcome was death within 30 days.

The result surprised many people who expected a benefit. According to the published summary, mortality at 30 days was 25.9 percent in the whole blood group and 20.5 percent in the component group, an adjusted odds ratio of 1.24 with a confidence interval running from 0.87 to 1.76, and a p-value of 0.24. In plain language, the trial did not show that whole blood helped, and the point estimate leaned the other way, though the uncertainty was wide enough that a benefit or a harm remained possible. The authors noted limitations: crossover between arms, variable volumes, an inability to mask the clinicians and limited data on coagulation. They also noted that overall mortality in the trial, 24.3 percent, was lower than in the earlier trial, which suggests that modern trauma systems have become better at controlling bleeding and delivering blood quickly.

I mention this trial for two reasons. First, it should humble anyone, including a BHOC advocate, who assumes that a physiologically attractive product will beat the alternative. Whole blood is about as physiologically attractive as a product can be, and in a rigorous test it did not win. Second, it shows what the bar looks like. A modern trauma system, delivering component therapy quickly, achieves a mortality figure around a fifth to a quarter in a severely injured population. A new product that hopes to demonstrate a mortality advantage in the same setting faces a very large sample size and a very narrow margin.

The three things a competitor cannot do

If the civilian trauma pathway is increasingly well served by blood, where is the gap? I would organize the answer around three functions that human blood products, however good, cannot perform. These are the defensible territory for a BHOC.

The first is to reach patients who will not accept blood. For patients whose conscience forbids transfusion, blood products are not competitors at all. They are unavailable by the patient's own decision. For those patients the relevant comparison is between a BHOC and no oxygen-carrying support other than what their own bodies can manage, which may mean life-threatening anemia treated with iron, erythropoietin, oxygen and hope. This is the setting in which emergency compassionate access has been most often sought, and it is the one where the ethical case is strongest. The population is small, but the need in each case is acute.

The second is to reach patients who cannot be matched. Some patients have developed antibodies against many red cell antigens, through repeated transfusions or pregnancies, so that compatible blood is hard or impossible to find. Some have rare blood types. In such a patient, a product with no red cell antigens and no need for cross-matching is not a marginal convenience but the only feasible option. The number of such patients is again modest, and their situation often unfolds slowly enough that the search for compatible blood can continue while a bridge is given.

The third is to survive where the cold chain cannot. Whole blood is stored at refrigerator temperature, and in the trial discussed above the shelf life was up to 21 days. Dried plasma travels better, but it supplies clotting factors and volume without red cells. Neither delivers oxygen-carrying capacity in a stable room-temperature form. A BHOC, on the other hand, has been described in product literature as storable at room temperature for years. There are places, remote clinics, disaster zones, small vessels at sea, long-range military patrols, where a cooler with a monitored temperature is a logistical burden that cannot be met. For those places, a shelf-stable oxygen carrier is not a substitute for blood so much as a way to keep a patient alive until blood can be reached.

The pack-mule arithmetic

Logistics is a matter of grams and degrees, and a short calculation shows why the room-temperature niche is more than sentiment. The figures are illustrative round numbers and do not describe any specific product or unit.

A medic on foot can carry perhaps a few kilograms of resuscitation supplies among everything else. A unit of refrigerated whole blood weighs roughly half a kilogram before packaging, and the insulated container and cold packs required to keep it between one and six degrees Celsius can multiply that several times over. The container must be opened rarely, monitored, and replaced when the ice runs out. A patrol that lasts three days in a hot climate has to plan for the ice.

A bag of a room-temperature carrier of comparable volume weighs about the same as the blood, without the container, and can ride in a pack beside the bandages. It also has no expiry that arrives in three weeks. If the patrol does not need it, it comes home and goes back on a shelf. Whole blood that was not used may be wasted or returned with its clock running down.

None of this makes the carrier better clinically. It makes it lighter and more forgiving, and in logistics that is sometimes the entire argument. Military planners talk about the burden of weight and the burden of refrigeration in the same breath, and any advocate who wants to be taken seriously by them will speak in those units.

The blood bank's view

It is instructive to imagine how a hospital blood bank director hears the BHOC pitch. Her whole professional life is organized around the safety of a product that is difficult to keep safe: matching, screening, storage, expiry, tracing. She is proud of the record, and she should be. She hears a proposal to add a new product to the shelf, made from another species, with a troubled history and a large question mark over its cardiac safety.

What would persuade her? Not a claim about the future of medicine. She would want four things. A defined list of patients for whom she can use it, so that her staff have a protocol and not a judgment call. A monitoring plan for the laboratory tests that a plasma pigment will disturb, so that chemistry results are not misread. A clear statement of who owns the decision to use it, and who documents consent. And a supply that does not run out, since a product that vanishes after she has written it into her protocols is worse than no product.

This is a useful test for a patient organization: if you cannot answer those four questions in a page, you are not ready to ask a blood bank director for anything. If you can, you are speaking her language, and she is likely to listen. Blood bankers are, for the most part, people who would like their patients to have more options, provided the options are safe and the paperwork is honest.

Where the pitch overreaches

It is just as useful to name the claims that a fair-minded clinician will not accept.

The claim that a BHOC can replace blood in general trauma is unsupported. The history of the field, including the pooled analysis published in 2008, gives reasons to be cautious, and the modern comparator has improved.

The claim that a BHOC is safer than blood because it lacks infectious risk is weaker than it was. The modern blood supply is screened for the known viruses, and the residual risks are very small. The bovine product introduces its own risks, addressed in another post in this series.

The claim that a BHOC is a universal solution to blood shortage is unrealistic. Shortages are real, but they are chronic and regional, and the oxygen-carrying capacity of a few units of a carrier is a small fraction of what a bleeding patient needs. A carrier has a half-life measured in hours to a day, and it does not replace clotting factors or platelets. A patient bleeding from a torn vessel needs the bleeding controlled and clotting support, and a BHOC provides neither.

That last point is worth dwelling on, because it changes the framing. Hemorrhage is a problem with three parts: stop the bleeding, restore clotting, restore oxygen delivery. The literature of the past two decades has shown that the first two have the greatest effect on survival, and that blood products address all three at once. A BHOC addresses only the third. It is a component, and a good use for it is as a bridge alongside other measures.

Combination, not competition

Here is a different way to see it. Suppose that the future of forward resuscitation lies in a combination of products chosen for their logistical properties. Dried plasma provides clotting factors and volume. A hemostatic agent controls bleeding. Whole blood, where cold chain exists, provides everything. And a shelf-stable oxygen carrier provides the oxygen-carrying capacity that plasma lacks when blood is unavailable.

The Army has published thinking about a lyophilized product strategy along these lines, combining freeze-dried plasma with a freeze-dried red cell or platelet component. That work is aimed at closing the gap between what a medic can carry and what a patient needs. In that architecture a BHOC is one of several possible oxygen components, and the question is whether it wins on weight, stability, cost and safety against the other candidates.

Stated that way, the comparison is more tractable and more honest. A BHOC does not have to beat blood. It has to beat the nearest alternative in a specific place. In a helicopter with a fridge, the alternative is blood. In a village three days' walk from a road, the alternative is nothing.

The evidence that would settle it

If a patient organization wanted to press the case in the sensible territory, what evidence would it ask for?

For the conscience-based population, a registry of every emergency access case, recorded consistently, with outcomes at discharge and follow-up, safety events, dose, hemoglobin measured both in plasma and in red cells, and the reasons for choosing the product. These cases are already occurring. Without a common form, the information is scattered across case reports.

For the unmatched population, a similar registry, together with a defined pathway under which a hospital blood bank can request the product when compatible blood cannot be found within a set time.

For the cold-chain-limited setting, a field-feasibility study: stock the product at remote sites, train staff, track storage conditions and use, and measure whether it is used appropriately and whether patients survive to reach definitive care. This is not a randomized trial, but it produces something regulators and funders need, which is proof that the logistics work.

For each, the outcome should be defined ahead of time and the comparator should be honest. Where the comparator is no oxygen-carrying therapy, a historical or registry control is acceptable if openly described. Where blood is available, the comparison should be to blood.

One more distinction protects everyone from confusion. Saying that a BHOC belongs where blood cannot go is different from saying that blood should not be used where it can. A patient who is bleeding in a well-equipped hospital should receive what the best evidence supports, which is blood in the right proportions. Nobody serves the field by suggesting otherwise, and nobody in the field who is honest with themselves believes it.

The strategic point

There is a temptation to answer the whole-blood trial by dismissing it, arguing that it was underpowered or that the population differed. The sounder response is to accept it and adjust. A field that has spent decades trying to beat blood in a general population has spent decades on the hardest version of the question. The easier version, and the one in which a BHOC has a clear reason to exist, is to reach the people that blood cannot reach.

That reframing also improves the ethics. A product aimed at patients who have no alternative has a favorable risk-benefit calculation even if its safety is imperfect, because the alternative is worse. A product aimed at patients who could receive blood must clear a higher bar. Being clear about which is which protects patients and protects the credibility of the advocates.

It also protects the science. If the only trials ever run are large trials against blood in trauma, the likely result is another disappointment, and the class will suffer for it. If the trials are designed around the populations with no alternative, the likely result is a clear signal, one way or the other, at a manageable sample size.

What I would say to a skeptical trauma surgeon

Imagine that a trauma surgeon who has read the recent trial asks why anyone still cares about a cow protein. A fair answer might go like this. You are right that in your system, with a helicopter, a blood bank and a cooler, the case for a carrier is weak. But you do not treat all the patients. Some refuse blood, some cannot be matched and some are in places your system does not reach. For those patients your options are thinner than you would like. A stable, unmatched, room-temperature product that keeps someone alive until blood is possible is worth having. I am not asking you to replace your cooler. I am asking you not to let the better answer for most patients become the enemy of a workable answer for the rest.

I think a surgeon might accept that. I think many would.

Closing

The simpler competitor has raised the bar, and in doing so it has clarified the mission. The story of the last fifteen years is not that oxygen carriers lost. It is that blood got better, and the space that remained is the space where blood cannot go. That space is smaller than the promoters once imagined, and it is real. Every person in it has a name and a family, and every one of them is a reason to do the work properly.

A cow protein will not rescue the general trauma patient. It might rescue the patient who has no other door. That is a modest ambition, and it happens to be the one the evidence can support.

Further reading

The prehospital plasma trial is Sperry and colleagues, New England Journal of Medicine, 2018 (PAMPer). The whole blood trial discussed here is TOWAR, published in the New England Journal of Medicine in 2026. A review of prehospital whole blood and plasma appeared in 2025, and the United States Army has published on a combined lyophilized product strategy. Numerical results quoted are from published summaries and abstracts and should be confirmed against the full reports.